Equations with variables on both sides

1 Explain what should be done to solve the equation $9x = 7 + 8x$.

2 Solve the following equations:

a $7x + 6 = 3x + 22$

b $6x - 3 = 4x + 7$

c $5x - 3 = -4x + 33$

d $4x - 9 = 5x - 6$

e $7x = x + 30$

f $3x = x - 18$

g $8x = 6x + 10$

h $7x = 3x - 12$

i $3x = -8x - 44$

j $10x + 8 = -8x - 2$

k $5x + 7 = 3x + 11$

l $11x + 15 = 6x + 35$

m $6x - 5 = 3x + 22$

n $-4x + 10 = 2x + 8$

o $16 + 4x = 12x$

p $18 - 2x = 9 + x$

q $24 - 6x = 14 + 4x$

r $30 - 10x = 22 - 8x$

3 Solve the following equations:

a $5x + 7 - 3x = x - 10$

b $8x + 6 - 2x = 4x + 8$

c $9x + 8 + 2x = 3x - 16$

d $10x + 12 - 2x = 6x + 8 - 4x$

4 Solve the following equations:

a $5\left(2x + 5\right) = 2x + 13$

b $6\left(3x + 1\right) = 4x + 20$

c $4\left(3x - 2\right) = 6x + 16$

d $2\left(8 - 3x\right) = 10x - 20$

e $3x + 4 = 5\left(6x + 5\right) + 60$

f $3x + 4 = 6\left(5x - 5\right) + 88$

g $4x - 6 = -8\left(6x - 7\right) + 42$

h $8\left(8x + 5\right) = 3\left(6x + 8\right) + 108$

i $5\left(4x - 3\right) = 3\left(-4x + 4\right) + 101$

j $-5\left(-5x + 5\right) = 7\left(4x - 8\right) + 37$

5 Solve the following equations:

a $2\left(3x + 4\right) + 5\left(x - 3\right) = 6\left(x + 10\right)$

b $5\left(2x + 1\right) - 3\left(3x - 1\right) = 2\left(x + 1\right)$

c $8\left(2x - 5\right) - 3x = 10 - 5\left(x + 9\right)$

d $2 - 6\left(2x + 1\right) = 4\left(3x - 1\right) + 6\left(x + 10\right)$

e $2\left(3x + 5\right) - \left(3x - 9\right) = 7 - 2\left(3x + 12\right)$

f $4\left(5x + 8\right) + 2\left(3x - 5\right) = 7\left(x + 11\right) - \left(2x - 8\right)$

6 Solve the following equations:

a $\dfrac{4x-40}{8}=3x$

b $\dfrac{4x+84}{8}=-3x$

c $\dfrac{-6x+54}{4}=3x$

d $\dfrac{3x+6}{7}=x+2$

e $\dfrac{-7x+7}{3}=6x+19$

f $\dfrac{5x-6}{7}=-5x+22$

g $\dfrac{4x+6}{6}=\dfrac{3x+12}{7}$

h $\dfrac{5x-4}{3}=\dfrac{3x+4}{5}$

i $\dfrac{4x+8}{8}=\dfrac{-3x+12}{3}$

j $\dfrac{8x-2}{9}=\dfrac{6x-4}{7}$

7 Consider the equation $3k+4=Ak-2$. If $k=6$ in this equation, find the value of A.

Applications

8 A number is multiplied by 5 and then 2 is added. Then the results is multiplied by 6. This is equal to 10 times the number minus 8.

a Form an equation for this problem.

b Solve the equation to find the number.

9 A rectangle with a height of $3x+7$ cm and a width of $4x$ has the same perimeter as a square with side length $2x+9$ cm.

Find the value of x.

10 A square with side length $4x-11$ cm has the same perimeter as a square with side length $2x+9$ cm.

Find the value of x.